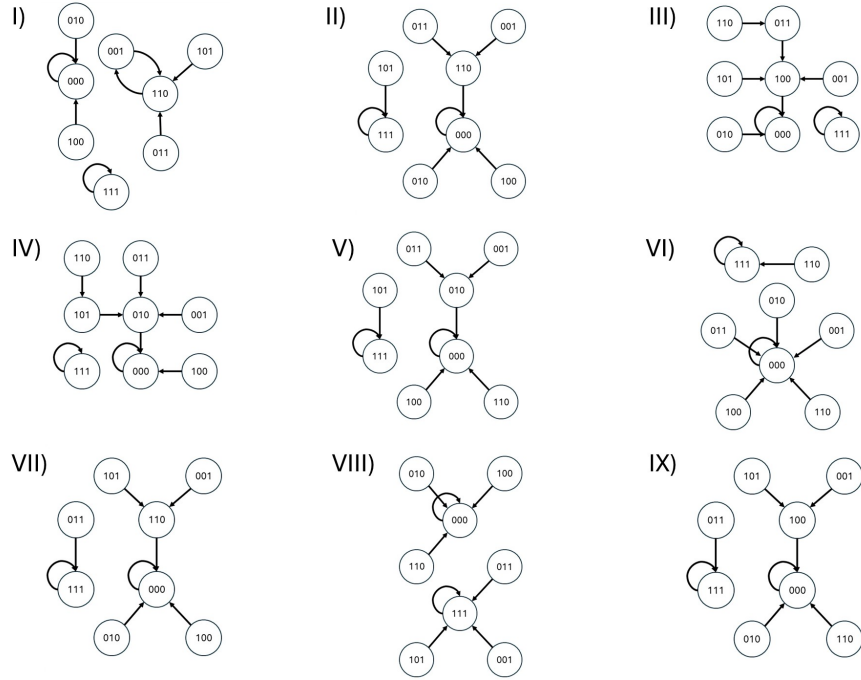


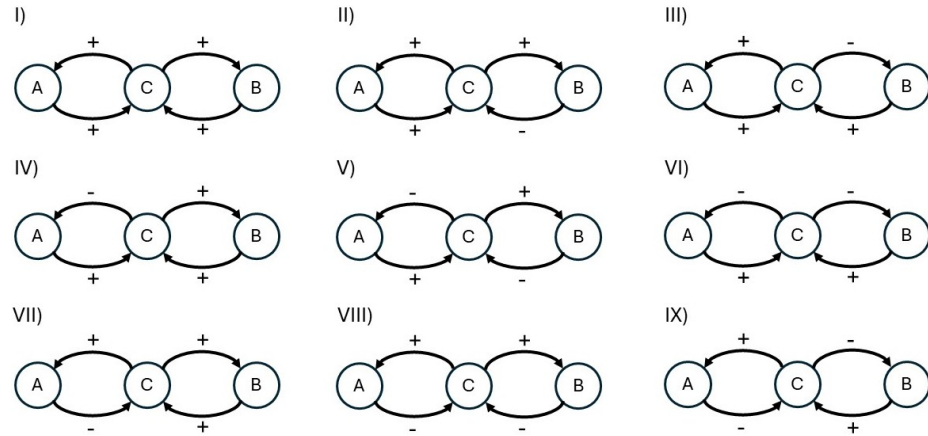
## 6 Online Supplementary Appendix

**Table S.1.** All the distinct dynamics (seen as transition tables) for the three-node example with  $V = \{A, B, C\}$ ,  $E = \{(A, C), (C, B), (B, C), (C, A)\}$ , and local rules  $x_A = x_C$ ,  $x_B = x_C$ ,  $x_C = x_A \wedge x_B$ , using the representative schemes  $s_1, \dots, s_9$ .

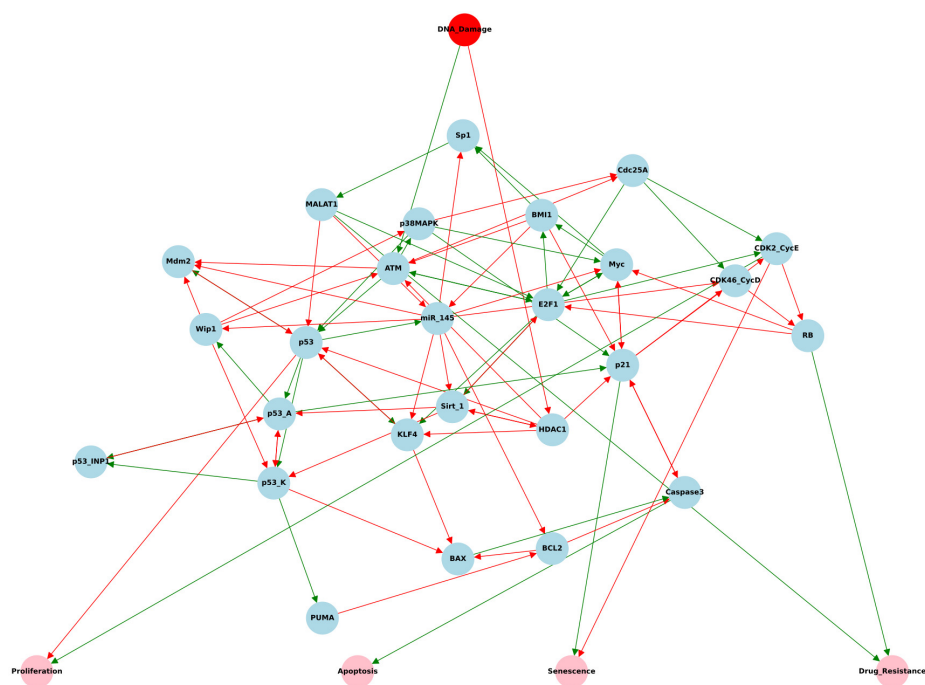
[illegible]



**Figure S.1.** All the distinct dynamics of Table S.1 seen as state transition graphs for the three-node update-digraph example, where I), ..., IX) are associated to the representative schemes  $s_1, \dots, s_9$ , respectively.



**Figure S.2.** All the update digraphs associated to the three-node example, where I) is obtained with  $s_1$ , II) with  $s_2$ , III) with  $s_3$ , IV) with  $s_4$ , V) with  $s_5$ , VI) with  $s_6$  (also with  $s_{10}$  and  $s_{13}$ ), VII) with  $s_7$ , VIII) with  $s_8$  (also with  $s_{11}$  and  $s_{12}$ ), and IX) with  $s_9$ .



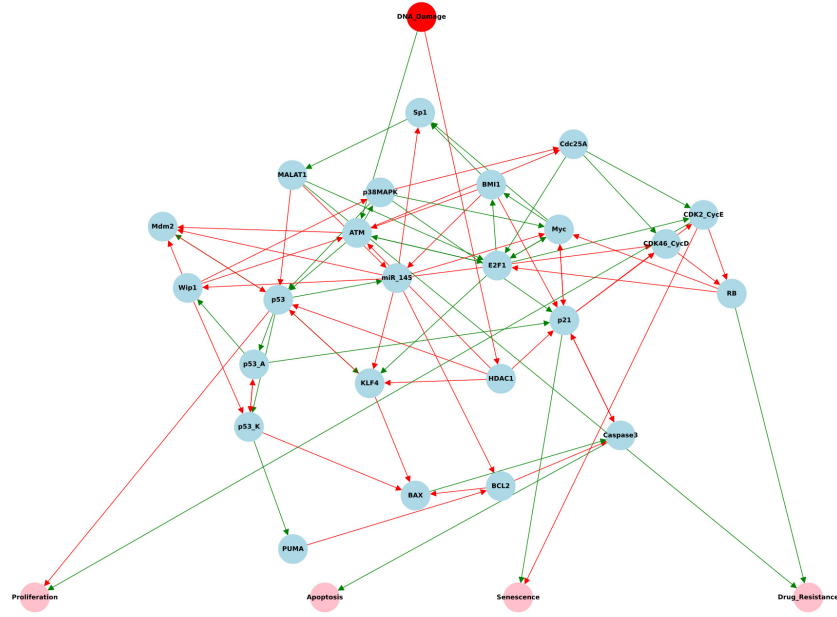
**Figure S.3.** The 31-node Boolean network for non-small cell lung cancer reconstructed from the one that appears in [2], where green arcs represent positive weights (activations), and red arcs represent negative weights (inhibitions). The red-colored node represents "DNA\_Damage", which corresponds to the input node; the four phenotypes that can be generated in this network are shown in pink, and the biological components corresponding to long non-coding RNAs, microRNAs, and various transcription factors are in blue.

**Table S.2.** The 30 regulated local boolean functions (or interactions) associated to the nodes of the NSCLC network shown in Figure S.3 (the input node DNA\_Damage is omitted).

| Node            | Interactions                                            |
|-----------------|---------------------------------------------------------|
| ATM             | DNA_Damage & (!HDAC1   !Wip1   E2F1   !BMI1)            |
| p38MAPK         | ATM & !Wip1                                             |
| miR_145         | p53 & !MALAT1 & !BMI1                                   |
| Sp1             | (BMI1 & Myc)   !miR_145                                 |
| MALAT1          | Sp1                                                     |
| BMI1            | Myc   E2F1                                              |
| KLF4            | !miR_145   (E2F1 & !HDAC1 & p53)                        |
| HDAC1           | !Sirt_1 & !DNA_Damage                                   |
| Myc             | (E2F1   p38MAPK   !p21) & !RB & !miR_145                |
| p53             | (ATM & !KLF4)   (!Mdm2 & p38MAPK & !HDAC1 & !MALAT1)    |
| Mdm2            | (!Wip1   p53) & !ATM & !miR_145                         |
| p53_A           | !Sirt_1 & !p53_K & (p53   !p53_INP1)                    |
| p53_K           | !p53_A & (!Sirt_1   !Wip1) & p53                        |
| Sirt_1          | E2F1 & !miR_145 & !HDAC1                                |
| p53_INP1        | p53_A   p53_K                                           |
| Wip1            | p53_A & !miR_145                                        |
| p21             | p53_A   ((!HDAC1 & !Myc) & !BMI1 & !Caspase3 & p38MAPK) |
| Cdc25A          | !ATM & !p38MAPK                                         |
| CDK46_CycD      | Cdc25A & !miR_145 & !p21                                |
| CDK2_CycE       | Cdc25A & E2F1 & !p21                                    |
| RB              | !CDK46_CycD & !CDK2_CycE                                |
| PUMA            | p53_K                                                   |
| BCL2            | !PUMA & !miR_145                                        |
| BAX             | (!BCL2   !KLF4) & !p53_K                                |
| E2F1            | (!RB & ((Cdc25A & ATM)   !Sirt_1))   MALAT1   Myc       |
| Caspase3        | (!BCL2   !p21) & BAX                                    |
| Proliferation   | CDK2_CycE & !p53                                        |
| Drug_Resistance | MALAT1 & RB                                             |
| Senescence      | p21 & !CDK2_CycE                                        |
| Apoptosis       | Caspase3                                                |

**Table S.3.** Steady state results for the 31-node network simulation.

| Components      | SteadyState1 | SteadyState2 | SteadyState3 | SteadyState4 | LimitCycle1 | LimitCycle2 | LimitCycle3 |
|-----------------|--------------|--------------|--------------|--------------|-------------|-------------|-------------|
| ATM             | 0            | 1            | 1            | 1            | 1 1         | 0 0 0 0     | 0 0 0 0     |
| p38MAPK         | 0            | 1            | 1            | 1            | 1 1         | 0 0 0 0     | 0 0 0 0     |
| miR_145         | 0            | 0            | 1            | 1            | 1 1         | 0 0 0 0     | 0 0 0 0     |
| Sp1             | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| MALAT1          | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| BMI1            | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| KLF4            | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| HDAC1           | 0            | 0            | 0            | 0            | 0 0         | 1 1 1 1     | 1 0 1 0     |
| Myc             | 1            | 0            | 0            | 0            | 0 0         | 1 1 0 0     | 1 1 0 1     |
| p53             | 0            | 0            | 1            | 1            | 1 1         | 0 0 0 0     | 0 0 0 0     |
| Mdm2            | 1            | 0            | 0            | 0            | 0 0         | 0 0 1 1     | 0 1 1 1     |
| p53_A           | 0            | 0            | 0            | 1            | 1 0         | 0 0 1 1     | 0 0 1 0     |
| p53_K           | 0            | 0            | 1            | 0            | 1 0         | 0 0 0 0     | 0 0 0 0     |
| Sirt_1          | 1            | 1            | 0            | 0            | 0 0         | 0 0 0 0     | 1 0 1 0     |
| p53_INP1        | 0            | 0            | 1            | 1            | 0 1         | 1 0 0 1     | 0 0 0 1     |
| Wip1            | 0            | 0            | 0            | 0            | 0 0         | 1 0 0 1     | 0 0 0 1     |
| p21             | 0            | 0            | 1            | 1            | 0 1         | 1 0 0 1     | 0 0 0 1     |
| Cdc25A          | 1            | 0            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| CDK46_CycD      | 1            | 0            | 0            | 0            | 0 0         | 0 0 1 1     | 0 1 1 1     |
| CDK2_CycE       | 1            | 0            | 0            | 0            | 0 0         | 0 0 1 1     | 0 1 1 1     |
| RB              | 0            | 1            | 1            | 1            | 1 1         | 0 1 1 0     | 0 1 0 0     |
| PUMA            | 0            | 0            | 1            | 0            | 0 1         | 0 0 0 0     | 0 0 0 0     |
| BCL2            | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| BAX             | 0            | 0            | 0            | 1            | 1 0         | 0 0 0 0     | 0 0 0 0     |
| E2F1            | 1            | 1            | 0            | 0            | 0 0         | 1 1 1 1     | 1 1 1 1     |
| Caspase3        | 0            | 0            | 0            | 1            | 0 1         | 0 0 0 0     | 0 0 0 0     |
| DNA_Damage      | 0            | 1            | 1            | 1            | 1 1         | 0 0 0 0     | 0 0 0 0     |
| Proliferation   | 1            | 0            | 0            | 0            | - -         | - - - -     | - - - -     |
| Drug Resistance | 0            | 1            | 0            | 0            | - -         | - - - -     | - - - -     |
| Senescence      | 0            | 0            | 1            | 1            | - -         | - - - -     | - - - -     |
| Apoptosis       | 0            | 0            | 0            | 1            | - -         | - - - -     | - - - -     |



**Figure S.4.** Reduced Boolean network for non-small cell lung cancer. For the construction of this network, the 31-node network was simplified with the most relevant genes and built based on the information in Table S.4, resulting in a total of 29 nodes. Green arcs represent positive weights (activations), and red arcs represent negative weights (inhibitions). The red-colored node represents “DNA\_Damage” which corresponds to the input node; the four phenotypes that can be generated in this network are shown in pink, and the biological components corresponding to long non-coding RNAs, microRNAs, and various transcription factors are in blue.

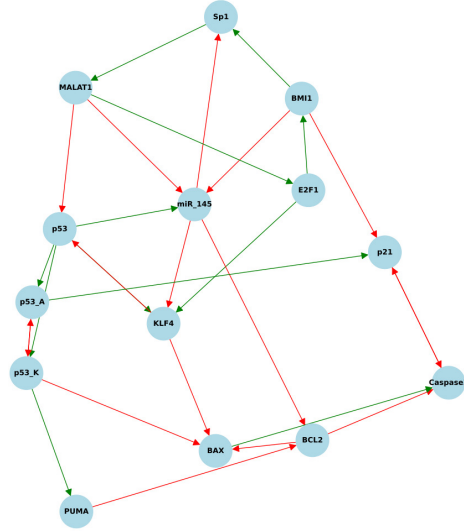
**Table S.4.** Adapted rules for the 29-node network.

| Node            | Interactions                                            |
|-----------------|---------------------------------------------------------|
| ATM             | DNA_Damage & (!HDAC1   !Wip1   E2F1   !BMI1)            |
| p38MAPK         | ATM & !Wip1                                             |
| miR_145         | p53 & !MALAT1 & !BMI1                                   |
| Sp1             | (BMI1 & Myc)   !miR_145                                 |
| MALAT1          | Sp1                                                     |
| BMI1            | Myc   E2F1                                              |
| KLF4            | !miR_145   (E2F1 & !HDAC1 & p53)                        |
| HDAC1           | !DNA_Damage                                             |
| Myc             | (E2F1   p38MAPK   !p21) & !RB & !miR_145                |
| p53             | (ATM & !KLF4)   (!Mdm2 & p38MAPK & !HDAC1 & !MALAT1)    |
| Mdm2            | (!Wip1   p53) & !ATM & !miR_145                         |
| p53_A           | !p53_K & (p53)                                          |
| p53_K           | !p53_A & (!Wip1) & p53                                  |
| Wip1            | p53_A & !miR_145                                        |
| p21             | p53_A   ((!HDAC1 & !Myc) & !BMI1 & !Caspase3 & p38MAPK) |
| Cdc25A          | !ATM & !p38MAPK                                         |
| CDK46_CycD      | Cdc25A & !miR_145 & !p21                                |
| CDK2_CycE       | Cdc25A & E2F1 & !p21                                    |
| RB              | !CDK46_CycD & !CDK2_CycE                                |
| PUMA            | p53_K                                                   |
| BCL2            | !PUMA & !miR_145                                        |
| BAX             | (!BCL2   !KLF4) & !p53_K                                |
| E2F1            | (!RB & ((Cdc25A & ATM)))   MALAT1   Myc                 |
| Caspase3        | (!BCL2   !p21) & BAX                                    |
| Proliferation   | CDK2_CycE & !p53                                        |
| Drug_Resistance | MALAT1 & RB                                             |
| Senescence      | p21 & !CDK2_CycE                                        |
| Apoptosis       | Caspase3                                                |

**Table S.5.** Steady state results for the 29-node network simulation.

| Components       | Steady state 1 | Steady state 2 | Steady state 3 | Steady state 4 | Limit Cycle 1 |
|------------------|----------------|----------------|----------------|----------------|---------------|
| ATM              | 0              | 1              | 1              | 1              | 1 1           |
| p38MAPK          | 0              | 1              | 1              | 1              | 1 1           |
| miR_145          | 0              | 0              | 1              | 1              | 1 1           |
| Sp1              | 1              | 1              | 0              | 0              | 0 0           |
| MALAT1           | 1              | 1              | 0              | 0              | 0 0           |
| BMI1             | 1              | 1              | 0              | 0              | 0 0           |
| KLF4             | 1              | 1              | 0              | 0              | 0 0           |
| HDAC1            | 0              | 0              | 0              | 0              | 0 0           |
| Myc              | 1              | 0              | 0              | 0              | 0 0           |
| p53              | 0              | 0              | 1              | 1              | 1 1           |
| Mdm2             | 1              | 0              | 0              | 0              | 0 0           |
| p53_A            | 0              | 0              | 0              | 1              | 1 0           |
| p53_K            | 0              | 0              | 1              | 0              | 1 0           |
| Wip1             | 0              | 0              | 0              | 0              | 0 0           |
| p21              | 0              | 0              | 1              | 1              | 0 1           |
| Cdc25A           | 1              | 0              | 0              | 0              | 0 0           |
| CDK46_CycD       | 1              | 0              | 0              | 0              | 0 0           |
| CDK2_CycE        | 1              | 0              | 0              | 0              | 0 0           |
| RB               | 0              | 1              | 1              | 1              | 1 1           |
| PUMA             | 0              | 0              | 1              | 0              | 0 1           |
| BCL2             | 1              | 1              | 0              | 0              | 0 0           |
| BAX              | 0              | 0              | 0              | 1              | 1 0           |
| E2F1             | 1              | 1              | 0              | 0              | 0 0           |
| Caspase3         | 0              | 0              | 0              | 1              | 0 1           |
| DNA_Damage       | 0              | 1              | 1              | 1              | 1 1           |
| Proliferation    | 1              | 0              | 0              | 0              | - -           |
| Drug Resistance  | 0              | 1              | 0              | 0              | - -           |
| Senescence       | 0              | 0              | 1              | 1              | - -           |
| Apoptosis        | 0              | 0              | 0              | 1              | - -           |
| Attraction basin | 50.00          | 49.89          | 0.01           | 0.04           | 0.05          |





**Figure S.5.** Reduced 14-node Boolean network for non-small cell lung cancer. To construct this network, the 29-node network was simplified from the information in Table S.5. The 29-node network was simplified for the case where “DNA\_Damage” equals 1, discarding nodes that fixed their configuration similarly in steady states and limit cycles, and also eliminating terminal nodes. Green arcs represent positive weights (activations), and red arcs represent negative weights (inhibitions).

**Table S.6.** Adapted rules for the 14-node network.

| Node     | Interactions                           |
|----------|----------------------------------------|
| miR_145  | $p53 \ \& \ !MALAT1 \ \& \ !BMI1$      |
| Sp1      | $(BMI1) \mid !miR\_145$                |
| MALAT1   | Sp1                                    |
| BMI1     | E2F1                                   |
| KLF4     | $!miR\_145 \mid (E2F1 \ \& \ p53)$     |
| p53      | $!KLF4 \mid !MALAT1$                   |
| p53_A    | $!p53\_K \ \& \ p53$                   |
| p53_K    | $!p53\_A \ \& \ p53$                   |
| BAX      | $(!BCL2 \mid !KLF4) \ \& \ !p53\_K$    |
| E2F1     | MALAT1                                 |
| Caspase3 | $(!BCL2 \mid !p21) \ \& \ BAX$         |
| p21      | $p53\_A \mid (!BMI1 \ \& \ !Caspase3)$ |
| BCL2     | $!PUMA \ \& \ !miR\_145$               |
| PUMA     | $p53\_K$                               |

**Table S.7.** Steady state results for the 14-node network simulation.

| Components       | Steady state 1 | Steady state 2 | Steady state 3 | Limit Cycle 1 |
|------------------|----------------|----------------|----------------|---------------|
| miR_145          | 0              | 1              | 1              | 1 1           |
| Sp1              | 1              | 0              | 0              | 0 0           |
| MALAT1           | 1              | 0              | 0              | 0 0           |
| BMI1             | 1              | 0              | 0              | 0 0           |
| KLF4             | 1              | 0              | 0              | 0 0           |
| p53              | 0              | 1              | 1              | 1 1           |
| p53_A            | 0              | 0              | 1              | 1 0           |
| p53_K            | 0              | 1              | 0              | 1 0           |
| p21              | 0              | 1              | 1              | 0 1           |
| PUMA             | 0              | 1              | 0              | 0 1           |
| BCL2             | 1              | 0              | 0              | 0 0           |
| BAX              | 0              | 0              | 1              | 1 0           |
| E2F1             | 1              | 0              | 0              | 0 0           |
| Caspase3         | 0              | 0              | 1              | 0 1           |
| Attraction basin | 98.44          | 0.39           | 0.39           | 0.78          |

**Table S.8.** Adapted rules for the 9-node network.

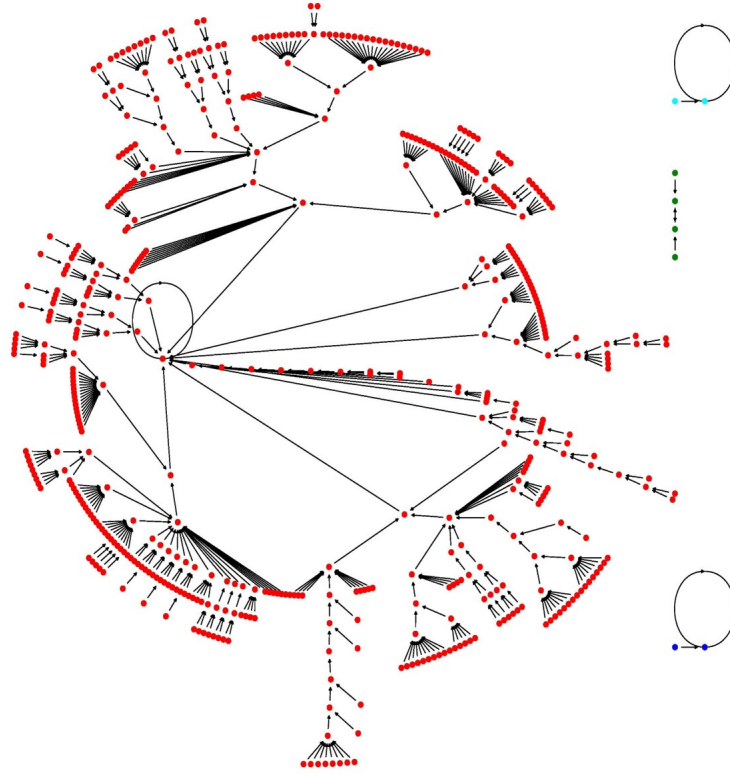
| Node    | Interactions            |
|---------|-------------------------|
| miR_145 | p53 & !MALAT1 & !BMI1   |
| Sp1     | (BMI1)   !miR_145       |
| MALAT1  | Sp1                     |
| BMI1    | E2F1                    |
| KLF4    | !miR_145   (E2F1 & p53) |
| p53     | !KLF4   !MALAT1         |
| p53_A   | !p53_K & p53            |
| p53_K   | !p53_A & p53            |
| E2F1    | MALAT1                  |

**Table S.9.** Limit cycle analysis for deterministic block-sequential update schemes of the minimal 9-node network.

| Steady state Configuration | Average Basin Attraction | SD    | Count  |
|----------------------------|--------------------------|-------|--------|
| 011110001                  | 441.19                   | 69.43 | 10,632 |
| 100001010                  | 23.25                    | 21.86 | 10,632 |
| 100001100                  | 23.25                    | 21.86 | 10,632 |

**Table S.10.** Top 10 limit cycles for deterministic block-sequential update schemes of the minimal 9-node network.

| Limit cycle Configuration | Average Basin Attraction | SD    | Count | Percent |
|---------------------------|--------------------------|-------|-------|---------|
| 100001000, 100001110      | 31.50                    | 26.57 | 2,658 | 67.29   |
| 010111101, 101001100      | 93.75                    | 57.41 | 32    | 0.81    |
| 010111100, 101001101      | 93.75                    | 57.41 | 32    | 0.81    |
| 010111011, 101001010      | 93.75                    | 57.41 | 32    | 0.81    |
| 010111010, 101001011      | 93.75                    | 57.41 | 32    | 0.81    |
| 010011100, 101101101      | 89.63                    | 61.81 | 32    | 0.81    |
| 010011010, 101101011      | 89.63                    | 61.81 | 32    | 0.81    |
| 001101101, 110011100      | 87.78                    | 53.75 | 32    | 0.81    |
| 001101011, 110011010      | 87.78                    | 53.75 | 32    | 0.81    |
| 000111101, 111001100      | 81.38                    | 44.17 | 32    | 0.81    |



**Figure S.6.** Attraction basin for the synchronous simulation of the 9-node network. This figure presents the attraction basins for the synchronous simulation of the 9-node network. Red points represent the attraction basin for “Drug Resistance”, with a size of 504 configurations. Light blue points represent the attraction basin for “Senescence”, with a size of 2 configurations. Blue points represent the attraction basin for “Apoptosis”, with a size of 2 configurations. Green points represent the attraction basin for the limit cycle, with a size of 4 configurations.

**Algorithm 1** FIT\_BOOLEAN\_RULES(Rules)

---

**Require:** Rules: dictionary mapping each node  $\rightarrow$  Boolean expression ( $\neg, \wedge, \vee$ )  
**Ensure:** ValidRules: dictionary mapping each node  $\rightarrow$  set of valid candidate rules

```

1: Nodes  $\leftarrow$  list of all nodes in Rules
   $\triangleright$  1) Compute original fixed points
2:  $A_{\text{original}} \leftarrow \emptyset$ 
3: for all state  $s \in \{0, 1\}^{|\text{Nodes}|}$  do
4:   if EvaluateNetwork(Rules,  $s$ ) =  $s$  then
5:      $A_{\text{original}} \leftarrow A_{\text{original}} \cup \{s\}$ 
6:   end if
7: end for
   $\triangleright$  2) Main loop over each target node
8: for all target  $\in$  Nodes do
9:   ValidRules[target]  $\leftarrow \emptyset$ 
10:  Inputs  $\leftarrow$  Nodes  $\setminus \{\text{target}\}$ 
   $\triangleright$  3) Generate input combinations of size 1–3
11:  for  $r = 1$  to 3 do
12:    for all subset  $C \subseteq$  Inputs with  $|C| = r$  do
   $\triangleright$  4) Generate candidate expressions for  $C$ 
13:    Candidates  $\leftarrow$  GENERATE_CANDIDATES( $C$ )
   $\triangleright$  5) Filter by functionality on fixed points
14:    for all expr  $\in$  Candidates do
15:      if  $\forall s \in A_{\text{original}} : \text{EVALUATE\_EXPR}(\text{expr}, s) = s[\text{target}]$  then
   $\triangleright$  6) Check attractor exactness
16:        Rules'  $\leftarrow$  copy of Rules
17:        Rules'[target]  $\leftarrow$  expr
18:         $A_{\text{new}} \leftarrow \emptyset$ 
19:        for all state  $t \in \{0, 1\}^{|\text{Nodes}|}$  do
20:          if EvaluateNetwork(Rules',  $t$ ) =  $t$  then
21:             $A_{\text{new}} \leftarrow A_{\text{new}} \cup \{t\}$ 
22:          end if
23:        end for
24:        if  $A_{\text{new}} = A_{\text{original}}$  then
25:          ValidRules[target]  $\leftarrow$  ValidRules[target]  $\cup \{\text{expr}\}$ 
26:        end if
27:      end if
28:    end for
29:  end for
30: end for
31: end for
32: return ValidRules

```

---

---

**Algorithm 2** GENERATE\_CANDIDATES( $C$ )

---

**Require:**  $C$ : set of  $r$  variables**Ensure:** **Candidates**: set of expressions

```

1: Candidates  $\leftarrow \emptyset$ 
2: if  $|C| = 1$  then
3:    $v \leftarrow$  the single element of  $C$ 
4:   Candidates  $\leftarrow \{v, \neg v\}$ 
5: else if  $|C| = 2$  then
6:    $\{x, y\} \leftarrow C$ 
7:   for all  $\text{sign\_x} \in \{\text{pos}, \text{neg}\}$  do
8:     for all  $\text{sign\_y} \in \{\text{pos}, \text{neg}\}$  do
9:        $\text{lit\_x} \leftarrow (\text{sign\_x} = \text{pos})?x : \neg x$ 
10:       $\text{lit\_y} \leftarrow (\text{sign\_y} = \text{pos})?y : \neg y$ 
11:      Candidates  $\leftarrow \text{Candidates} \cup \{\text{lit\_x} \wedge \text{lit\_y}, \text{lit\_x} \vee \text{lit\_y}\}$ 
12:    end for
13:  end for
14: else if  $|C| = 3$  then
15:    $\{X, Y, Z\} \leftarrow C$ 
16:   for all each assignment of negation flags over  $\{X, Y, Z\}$  do
17:     define  $\text{lit\_X}, \text{lit\_Y}, \text{lit\_Z}$ 
18:     Candidates  $\leftarrow \text{Candidates} \cup \{\text{lit\_X} \wedge \text{lit\_Y} \wedge \text{lit\_Z}, \text{lit\_X} \vee \text{lit\_Y} \vee \text{lit\_Z}\}$ 
19:     Candidates  $\leftarrow \text{Candidates} \cup \{(\text{lit\_X} \wedge \text{lit\_Y}) \vee \text{lit\_Z}, (\text{lit\_X} \vee \text{lit\_Y}) \wedge$ 
       $\text{lit\_Z}\}$ 
20:   end for
21: end if
22: return Candidates

```

---



---

**Algorithm 3** EvaluateNetwork(**Rules**,  $s$ )

---

 $\triangleright$  Synchronous update: evaluate each rule with state  $s$ , return new state  $s'$ .

---

---

**Algorithm 4** EVALUATE\_EXPR( $\text{expr}, s$ )

---

 $\triangleright$  Substitute literals with values from  $s$  and evaluate  $\rightarrow$  Boolean.

---

**Table S.11.** fitted rules for the 9-node network.

| Node    | Interactions             |
|---------|--------------------------|
| miR_145 | p53 & !MALAT1 & !BMI1    |
| Sp1     | (BMI1)   !miR_145        |
| MALAT1  | Sp1                      |
| BMI1    | (!p53_A & !p53_K)   E2F1 |
| KLF4    | !miR_145   (E2F1 & p53)  |
| p53     | !KLF4   !MALAT1          |
| p53_A   | !p53_K & p53             |
| p53_K   | !p53_A & p53             |
| E2F1    | MALAT1                   |

**Table S.12.** Steady state results for the fitted 9-node network simulation.

| Components       | Steady state 1 | Steady state 2 | Steady state 3 |
|------------------|----------------|----------------|----------------|
| miR_145          | 0              | 1              | 1              |
| Sp1              | 1              | 0              | 0              |
| MALAT1           | 1              | 0              | 0              |
| BMI1             | 1              | 0              | 0              |
| KLF4             | 1              | 0              | 0              |
| p53              | 0              | 1              | 1              |
| p53_A            | 0              | 0              | 1              |
| p53_K            | 0              | 1              | 0              |
| E2F1             | 1              | 0              | 0              |
| Attraction basin | 99.22          | 0.39           | 0.39           |

**Table S.13.** Limit cycle analysis for deterministic block-sequential update schemes of the fitted 9-node network simulation.

| Steady state Configuration | Average Basin Attraction | SD    | Count  |
|----------------------------|--------------------------|-------|--------|
| 011110001                  | 460.87                   | 54.98 | 23,107 |
| 100001010                  | 22.47                    | 18.46 | 23,107 |
| 100001100                  | 22.47                    | 18.46 | 23,107 |

**Table S.14.** Top 10 limit cycles for deterministic block-sequential update schemes of the fitted 9-node network simulation.

| Limit cycle Configuration | Average Basin Attraction | SD    | Count | Percent |
|---------------------------|--------------------------|-------|-------|---------|
| 010011100, 101101101      | 78.87                    | 43.28 | 67    | 3.28    |
| 010011010, 101101011      | 78.87                    | 43.28 | 67    | 3.28    |
| 000111100, 111001101      | 60.42                    | 32.22 | 67    | 3.28    |
| 000111010, 111001011      | 60.42                    | 32.22 | 67    | 3.28    |
| 010111010, 101001011      | 88.36                    | 52.96 | 67    | 3.28    |
| 010111100, 101001101      | 88.36                    | 52.96 | 67    | 3.28    |
| 000111101, 111001100      | 77.01                    | 39.66 | 67    | 3.28    |
| 000111011, 111001010      | 77.01                    | 39.66 | 67    | 3.28    |
| 010111011, 101001010      | 89.07                    | 53.82 | 67    | 3.28    |
| 010111101, 101001100      | 89.07                    | 53.82 | 67    | 3.28    |